

WORKSHOP PRACTICES

B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A4AE63	ESC	L	T	P	C	CIE	SEE	Total
		-	-	4	2	30	70	100

COURSE OBJECTIVES:

Student will

1. Get the hands-on experience on various trades.
2. Perform various machining operations.
3. Capable to make useful products using one or more operations

COURSE OUTCOMES:

Student will be able to:

1. Fabricate components with their own hands
2. Get practical knowledge of the dimensional accuracies and tolerances possible with different manufacturing processes
3. Assemble different components
4. Produce small devices of their interest

LIST OF EXPERIMENTS

WEEKS	BASIC TRADES	BASIC MANUFACTURING
	Fitting	Machine Shop
WEEK 1	Filing Four Sides of Work piece	Facing & Step Turning on Lathe
WEEK 2	L- Fit	Milling and Drilling
	Carpentry	Black Smithy
WEEK 3	Half Lap Joint	Convert round rod to S-hook
WEEK 4	Dove Tail Joint	Convert round rod to Chisel
	Tin Smithy	Casting
WEEK 5	Tin Smithy- Prepare a Rectangular Tray	Preparation of Mould Cavity for Multi Piece Pattern
WEEK 6	Prepare A Square Tin	Casting of Simple pattern
	Electrical	Welding Shop
WEEK 7	House Wiring Parallel and Series Connection	Lap/Butt joint Using Arc Welding
WEEK 8	House Wiring Two Way Switch	Lap/Butt joint Using Gas Welding
	Electronics	Plastic Moulding & Glass Cutting
WEEK 9	Soldering Parallel Connection	Injection moulding of Simple Components
WEEK 10	Soldering Series Connection	Glass Cutting
WEEK 11	Revision/Practice	Revision/Practice

TEXT BOOKS:

1. Workshop Manual by P. Kannaiah and K. L. Narayana.
2. Rao P.N., "Manufacturing Technology", Tata McGraw Hill House, Vol. I and Vol. II.

REFERENCE BOOKS:

1. Hajra Choudhury S.K., Hajra Choudhury A.K. and Nirjhar Roy S.K., "Elements of Workshop Technology", Media promoters and publishers private limited, Mumbai, Vol. I 2008 and Vol. II 2010.
2. H. S. Bawa, "Workshop Practice", Tata McGraw-Hill Publishing Company Limited, New Delhi, (2007).
3. Kalpakjian S. and Steven S. Schmid, "Manufacturing Engineering and Technology", Pearson Education India Edition, 4th edition, 2002.

E-TEXT BOOKS:

1. <https://blogpuneet.files.wordpress.com/2013/07/introduction-to-basic-manufacturing-processes-and-workshop-technology.pdf>
2. <https://soaneemrana.org/onewebmedia/Manufacturing%20Processes%20By%20H.N.%20Gupta.pdf>

MOOC COURSE:

1. <https://www.class-central.com/course/edx-fundamentals-of-manufacturing-processes-7224>